- the construction of a pumping station to flush the creek and input oxygen into the dead water body
- the installation of a water lock between the Huangpu River and Suzhou Creek
- the construction of a wastewater treatment plant, with a capacity of 400,000 m^3 a day
- the installation of solid waste collection wharves.

The second phase followed on immediately from the first phase and was completed in 2005. Its objectives were:

- to maintain and improve the water quality (building on the improvements in the first phase)
- to extend the cleaning to its six tributaries
- to construct embankments to develop large areas of green space along the river banks.

The third phase (2006–08) continued the emphasis on improving riverside accessibility for surrounding residents. Public use of riverfront space, appropriate distribution of a riverfront greenbelt and regulation of riverfront buildings were included in the Suzhou Creek Landscape Planning. This final phase emphasised the social and environmental benefits of the project to local residents.

The combination of these measures removed wastewater from the creek and kept it out. Overall, the project benefited about 3 million people living in proximity to the creek. It improved the environment of a considerable area of Shanghai by:

- improving sanitation services
- reducing serious risks to public health
- providing greater access to parks and green spaces along the river banks.

Public surveys conducted after completion of the project showed that satisfaction with the city's environment had increased from 12 per cent in 2000 to 71 per cent in 2003. Satisfaction with water quality had risen from 12 per cent in 2000 to 76 per cent in 2003.

Finance and project partners

The initiation and ultimate success of the project depended on collaboration between a number of partner organisations. The total cost of the project was US$876 million, comprised of:

- a US$300 million loan from the Asian Development Bank
- US$325.4 million from the State Development Bank
- US$132.5 million from the Ministry of Finance
- US$62.7 million from the Shanghai Municipal Government
- US$55.4 million from district and county governments.

Sustainable management

The Chinese government has stressed environmental protection and sustainable management as a national strategy under its 'Ecological Civilisation' programme. UN HABITAT has also recognised the regeneration of Suzhou Creek as a success.

Major sustainability measures include:

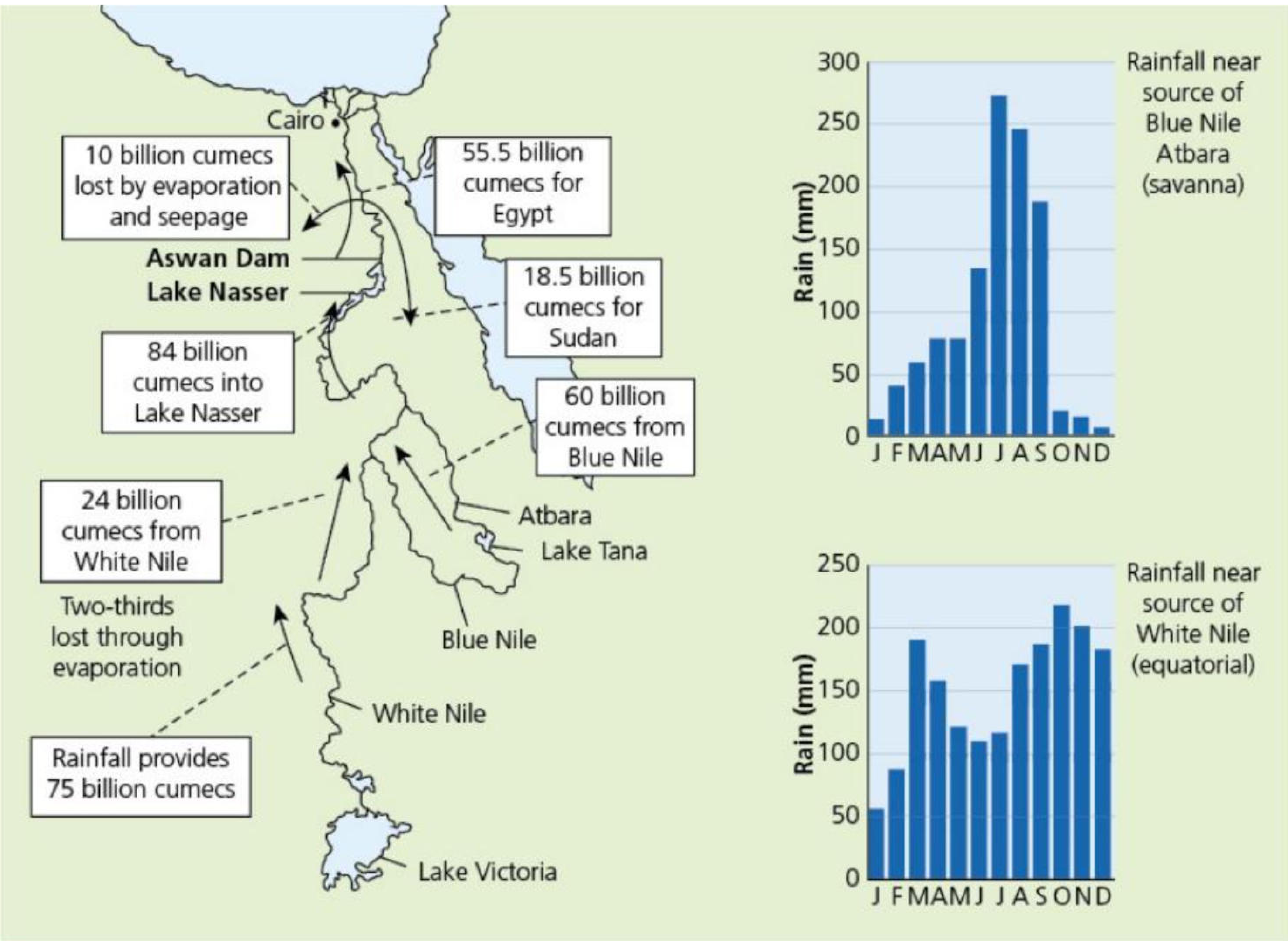
- Education and legislation to reduce pollutants entering the river system.
- A 'River Chief' system of responsibility at all levels of government has been established for the management and protection of rivers.
- Maintaining a sufficient level of investment to monitor water quality and maintain the necessary quality of infrastructure. Continuous monitoring of pollution levels in the drainage basin as a whole is vitally important for successful sustainability.

Activities

1. Why is Suzhou Creek regarded as an important waterway?
2. How did the river become so polluted?
3. Describe the three phases in the restoration of the river.
4. Suggest why so many organisations were involved in the restoration project.

Practice questions

1. Figure 1.47 shows the inflows and outflows on the Nile water and rainfall graphs for the source areas.

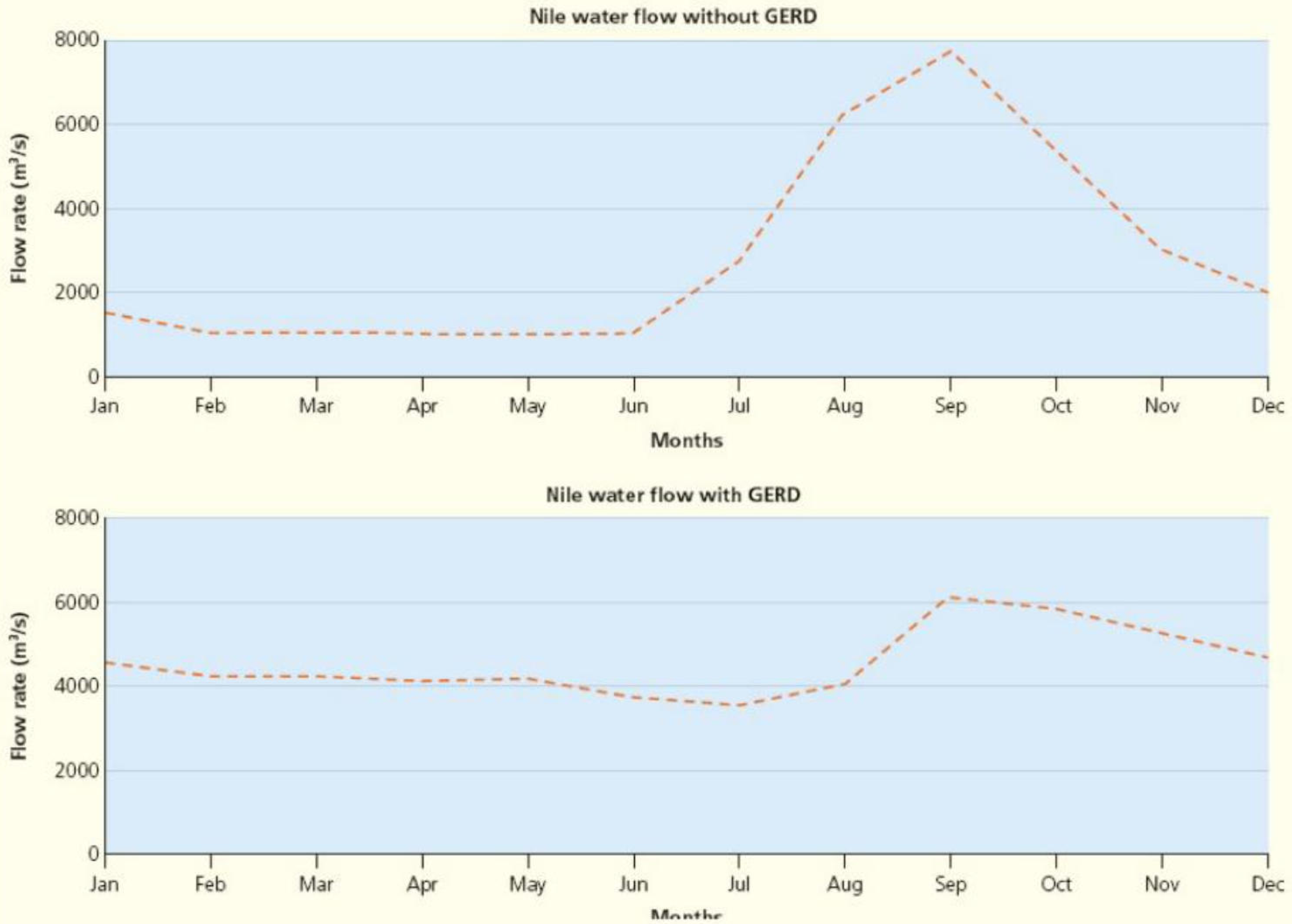


▲ **Figure 1.47** The inflows and outflows on the Nile water and rainfall graphs for the source areas

- **a** State the amount of water that:
  - **i** initially comes from the White Nile [1]
  - **ii** joins the Blue Nile. [1]
- **b** Describe the rainfall pattern for the source of the Blue Nile. [3]

2 Figure 1.48 shows the changes in the flow of water in the Nile before and after the building of the Grand Ethiopian Renaissance Dam (GERD), completed in 2020.

- **a** State the maximum flow in the Nile:
  - **i** without the GERD [1]
  - **ii** with the GERD. [1]
- **b** Compare the flow of the Blue Nile without the GERD and with the GERD. [3]



▲ **Figure 1.48** The changes in the flow of water in the Nile before and after the building of the GERD, completed in 2020